

 **Important**
Use Kiro **CLI** for this section

Prerequisites

 **Note**
If you haven't finished [Enhancing the UI with Detailed Prompts](#)

Run the following prompt to create Sudoku game that will be deployed in this lab.

```
1 I want to create a simple html Sudoku game UI with the following features:
2 1. A modern, clean design with a color scheme that reduces eye strain
3 2. A responsive layout that works well on both desktop and mobile devices
4 3. Visual feedback for selected cells, valid/invalid entries, and completed rows/columns/boxes
5 4. A game info panel showing:
6   - Current difficulty level
7   - Timer
8   - Number of mistakes
9   - Completion percentage
10 5. Animations for cell selection and number placement
11 6. Support for both mouse and keyboard input
```



1. Create Infrastructure as Code (IaC) -> AWS Cloud Development Kit (CDK)

```
1 Use AWS Cloud Development Kit (CDK) to implement infrastructure as code in Python.
2 Using the following AWS solutions:
3 - Host the web application in Amazon S3 private bucket
4 - Deliver the web content through Amazon CloudFront.
```



2. Verify output

Kiro will create the CDK code and provide instructions to deploy to AWS

LAB-0

> cdk-sudoku-app

<> index.html

📘 README.md

JS script.js

style.css

> OUTLINE

> TIMELINE

> APPLICATION BUILDER

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

CODE REFERENCE LOG

4. Run the deployment script:

```
bash
python deploy_website.py
```

This will deploy the infrastructure and upload your Sudoku web application files to the S3 bucket. Once complete, you'll receive the CloudFront URL where your application is accessible.

Benefits of This Approach

1. **Security:** The S3 bucket is private, and content is only accessible through CloudFront

2. **Performance:** CloudFront provides global content delivery with low latency

3. **Cost-Effective:** Pay only for the storage and data transfer you use

4. **Infrastructure as Code:** The entire setup is defined as code, making it reproducible and maintainable

5. **Scalability:** CloudFront automatically scales to handle any traffic load

Would you like me to explain any specific part of the implementation in more detail?

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